14. Main Group Chemistry—Boranes, borazines, phosphazenes and cyclic phosphazene, silicates and silicones, Interhalogen compounds; Sulphur–nitrogen compounds, noble gas compounds.

15. General Chemistry of 'f' Block Elements—Lanthanides and actinides; separation, oxidation states, magnetic and spectral properties; lanthanide contraction.

PAPER–II

1. Delocalised Covalent Bonding—Aromaticity, anti-aromaticity; annulenes, azulenes, tropolones, fulvenes, sydnones.

2. (i) Reaction Mechanisms—General methods (both kinetic and non-kinetic) of study of mechanism of organic reactions: isotopic method, cross-over experiment, intermediate trapping, stereochemistry; energy of activation; thermodynamic control and kinetic control of reactions.

(ii) Reactive Intermediates—Generation, geometry, stability and reactions of carbonium ions and carbanions, free radicals, carbenes, benzynes and nitrenes.

(iii) Substitution Reactions—S_N1, S_N2 and S_Ni mechanisms; neighbouring group participation; electrophilic and nucleophilic reactions of aromatic compounds including heterocyclic compounds–pyrrole, furan, thiophene and indole.

(iv) Elimination Reactions—E1, E2 and E1cb mechanisms; orientation in E2 reactions–Saytzeff and Hoffmann; pyrolytic **syn** elimination–Chugaev and Cope eliminations.

(v) Addition Reactions—Electrophilic addition to C = C and C ≡ C; nucleophilic addition to C = 0, C ≡ N, conjugated olefins and carbonyls.

(vi) Reactions and Rearrangements—(a) Pinacol-pinacolone, Hoffmann, Beckmann, Baeyer–Villiger, Favorskii, Fries, Claisen, Cope, Stevens and Wagner-Meerwein rearrangements.

(b) Aldol condensation, Claisen condensation, Dieckmann, Perkin, Knoevenagel, Witting, Clemmensen, Wolff-Kishner, Cannizzaro and von Richter reactions; Stobbe, benzoin and acyloin condensations; Fischer indole synthesis, Skraup synthesis, Bischler-Napieralski, Sandmeyer, Reimer-Tiemann and Reformatsky reactions.

3. Pericyclic Reactions—Classification and examples; Woodward-Hoffmann rules–electrocyclic reactions, cycloaddition reactions [2+2 and 4+2] and sigmatropic shifts [1, 3; 3, 3 and 1, 5] FMO approach.

4. (i) Preparation and Properties of Polymers—Organic polymers–polyethylene, polystyrene, polyvinyl chloride, teflon, nylon, terylene, synthetic and natural rubber.

(ii) Biopolymers: Structure of proteins, DNA and RNA.

5. Synthetic Uses of Reagents—OsO_4, HIO_4, CrO_3, $Pb(OAc)_4$, SeO_2, NBS, B_2H_6, Na-Liquid NH_3, $LiAlH_4$, $NaBH_4$, n-BuLi and MCPBA.

6. Photochemistry—Photochemical reactions of simple organic compounds, excited and ground states, singlet and triplet states, Norrish-Type I and Type II reactions.

7. Spectroscopy—Principle and applications in structure elucidation:

(i) Rotational: Diatomic molecules; isotopic substitution and rotational constants.

(ii) Vibrational—Diatomic molecules, linear triatomic molecules, specific frequencies of functional groups in polyatomic molecules.

(iii) Electronic—Singlet and triplet states; $n \rightarrow \pi^*$ and $\pi\pi^* \rightarrow$ transitions; application to conjugated double bonds and conjugated carbonyls–Woodward-Fieser rules; Charge transfer spectra.

(iv) Nuclear Magnetic Resonance (^{1}H NMR)—Basic principle; chemical shift and spin-spin interaction and coupling constants.

(v) Mass Spectrometry—Parent peak, base peak, metastable peak, McLafferty rearrangement.

CIVIL ENGINEERING

PAPER–I

1. Engineering Mechanics, Strength of Materials and Structural Analysis :

1.1 Engineering Mechanics—Units and Dimensions, SI Units, Vectors, concept of Force, Concept of particle and rigid body. Concurrent, Non-concurrent and parallel forces in a plane, moment of force, free body diagram, conditions of equilibrium, Principle of virtual work, equivalent force system.

First and Second Moment of area, Mass moment of Inertia.

Static Friction, Kinematics and Kinetics :

Kinematics in Cartesian Co-ordinates, motion under uniform and nonuniform acceleration, motion under gravity. Kinetics of particle: Momentum and Energy principles, collision of elastic bodies, rotation of rigid bodies.

1.2 Strength of Materials—Simple Stress and Strain, Elastic constants, axially loaded compression members, Shear force and bending moment, theory of simple bending, Shear Stress distribution across cross sections, Beams of uniform strength.